

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 994+520

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – VUP

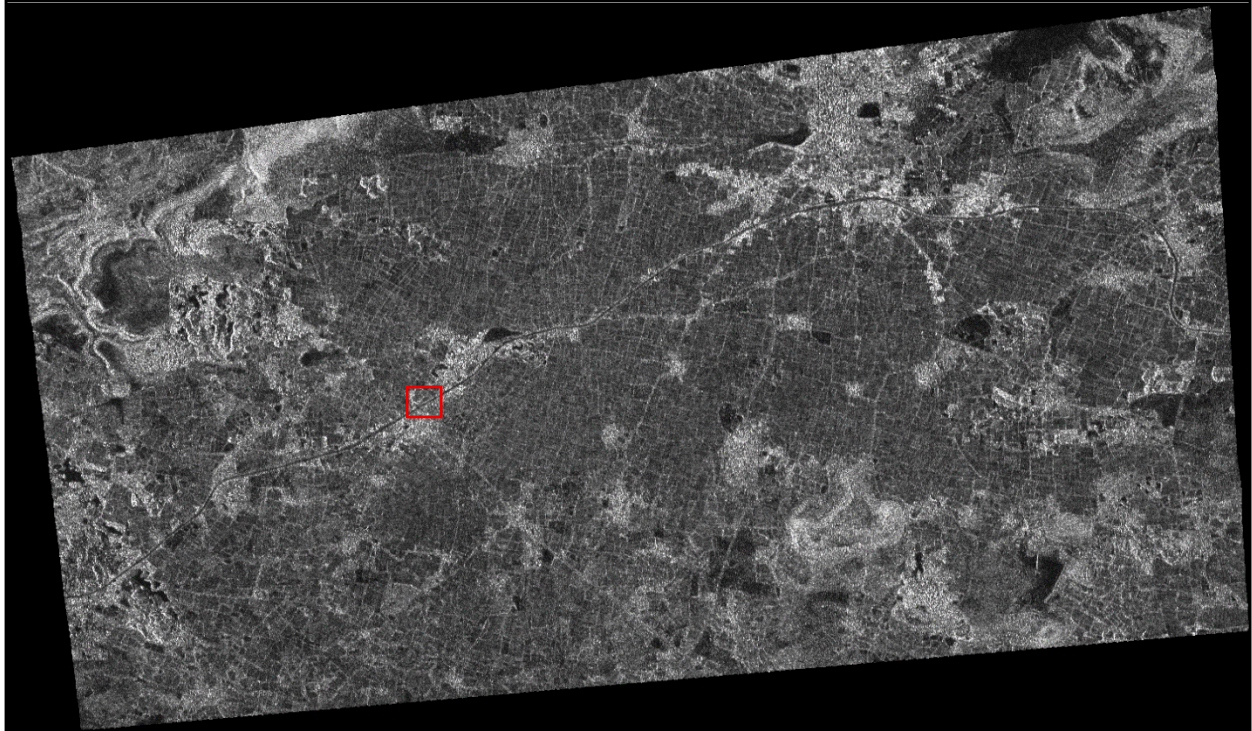
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

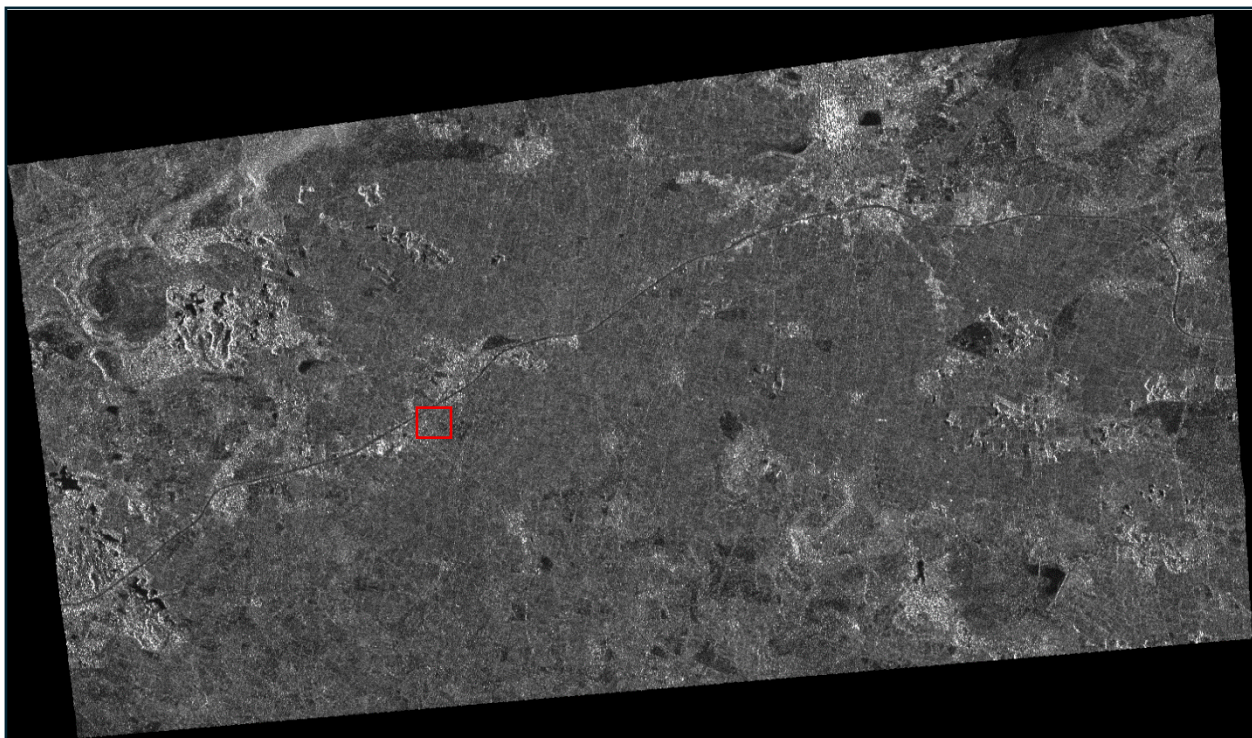
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

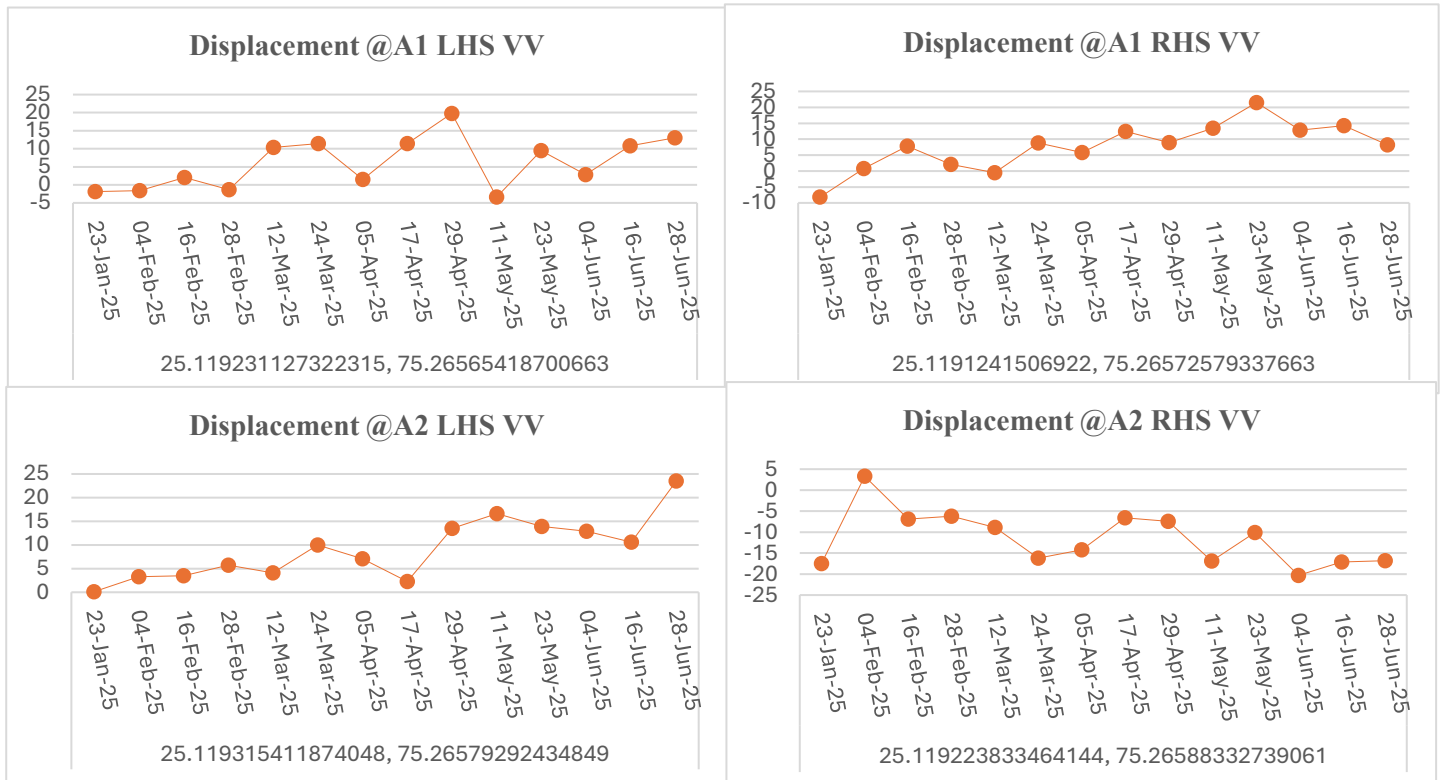
VUP – CH-994+520

Length – 32.94M

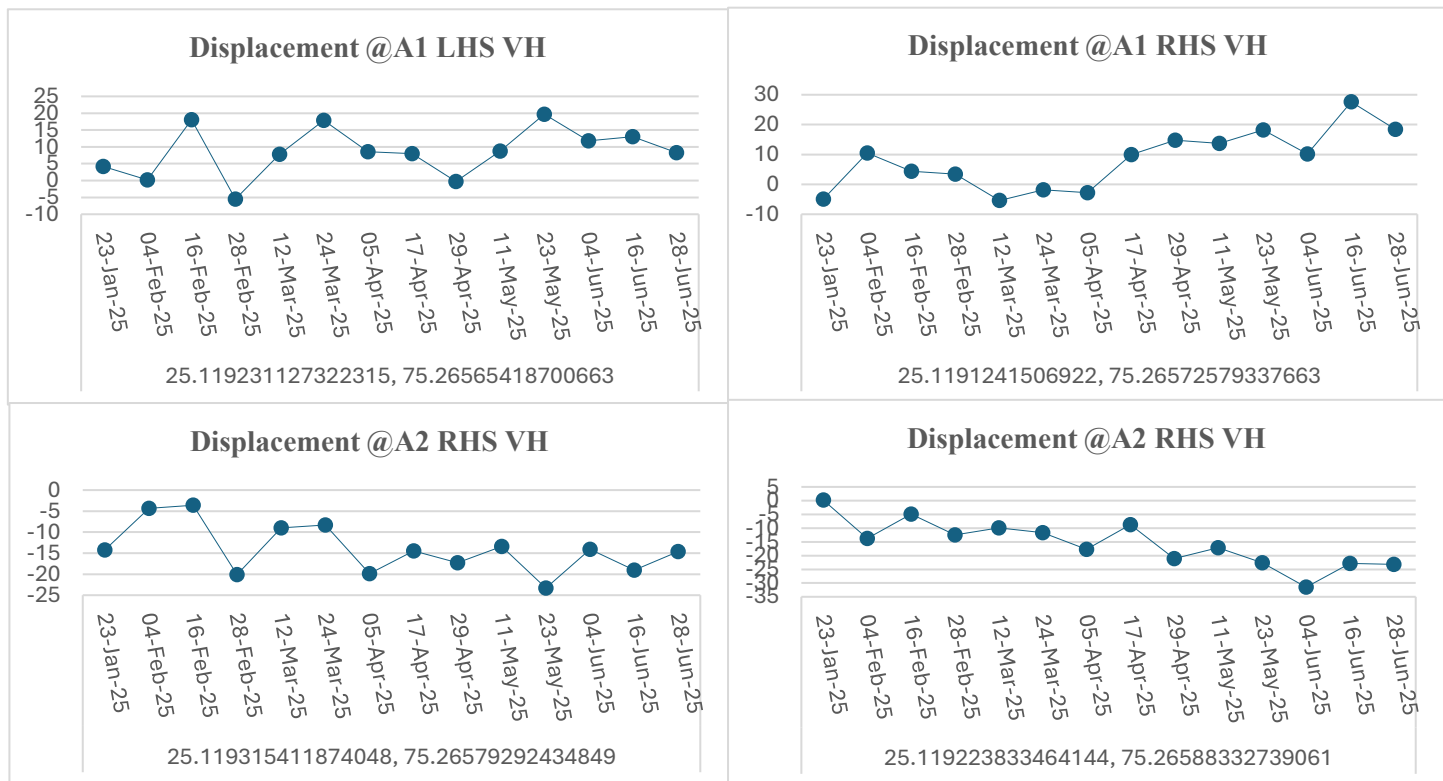


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



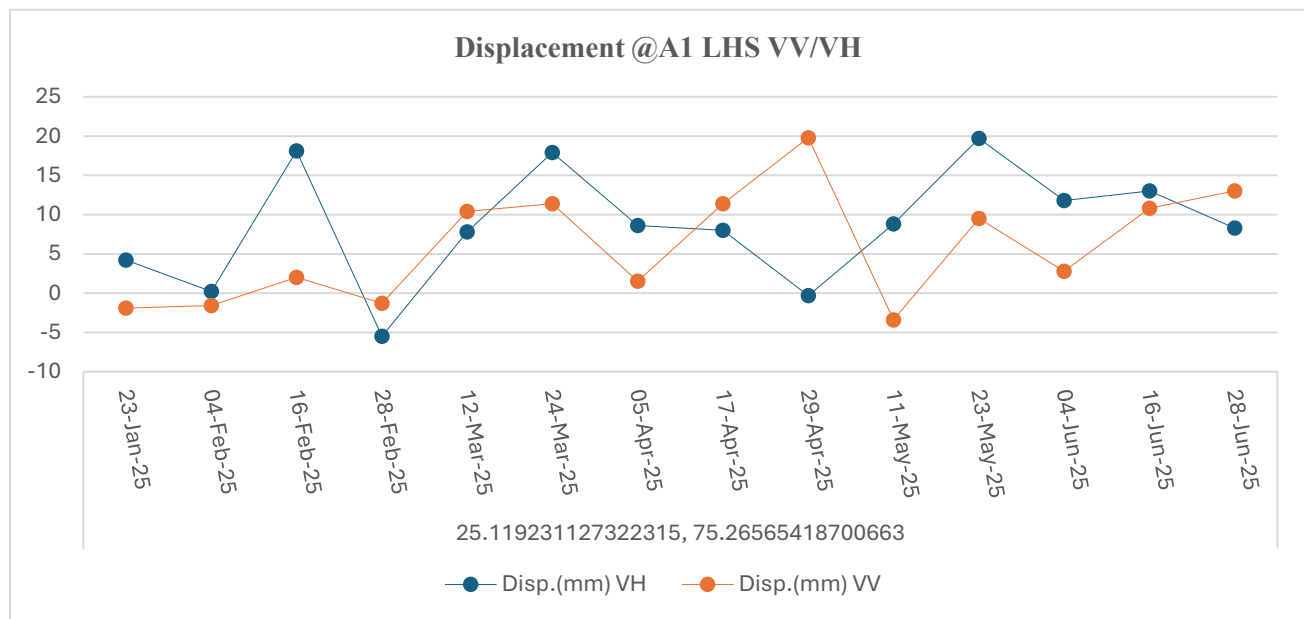
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.119231127322315, 75.26565418700663)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate-to-high variability, fluctuating between approximately -4 mm (settlement) and +20 mm (uplift).
- **Horizontal Displacement (VH)** Horizontal displacement at A1 LHS shows high variability with multiple sharp peaks, ranging approximately from -5 mm to +19 mm.

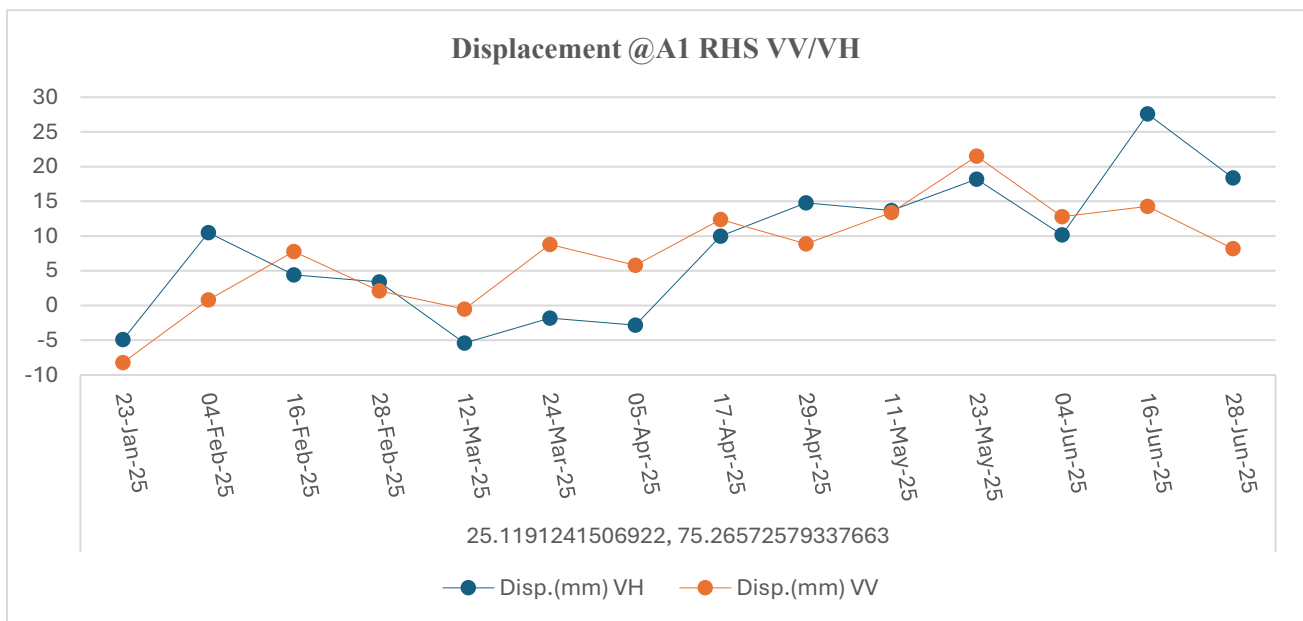
Critical Events & Anomalies:

- **January 2025:** VH begins at a moderate positive value, while VV is slightly negative, indicating initial lateral displacement with minor vertical settlement.
- **June 2025:** Both VH and VV remain positive and moderate, with VH around +8–13 mm and VV around +10–13 mm, indicating stabilisation with sustained uplift and lateral displacement toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.1191241506922, 75.26572579337663)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A1 RHS shows high variability with a clear increasing trend over time, ranging approximately from -5 mm to +27 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits moderate variability, ranging approximately between -8 mm (settlement) and +21 mm (uplift).

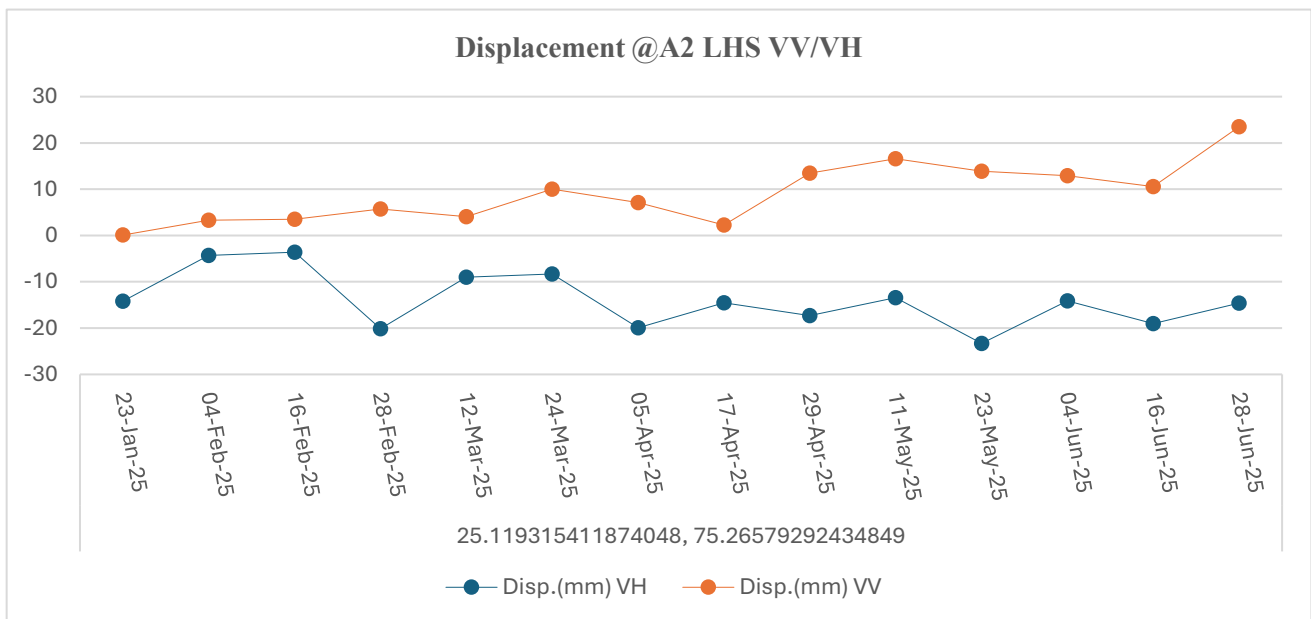
Critical Events & Anomalies:

- **January-February 2025:** VH starts slightly negative (-5 mm), while VV is more negative (-9 mm), indicating initial combined lateral and vertical settlement at this location.
- **May 2025:** VV peaks at approximately +21 mm, while VH also remains high (18 mm), marking the maximum combined displacement event in the monitoring period.
- **June 2025:** VH reaches its absolute maximum (+27 mm), while VV decreases to 14 mm, indicating continued lateral progression with reduced vertical contribution.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.119315411874048, 75.26579292434849)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS remains consistently negative throughout the monitoring period, indicating sustained lateral movement in one direction.
- **Vertical Displacement (VV)** exhibits moderate short-term fluctuations but maintains a dominant upward progression over time.

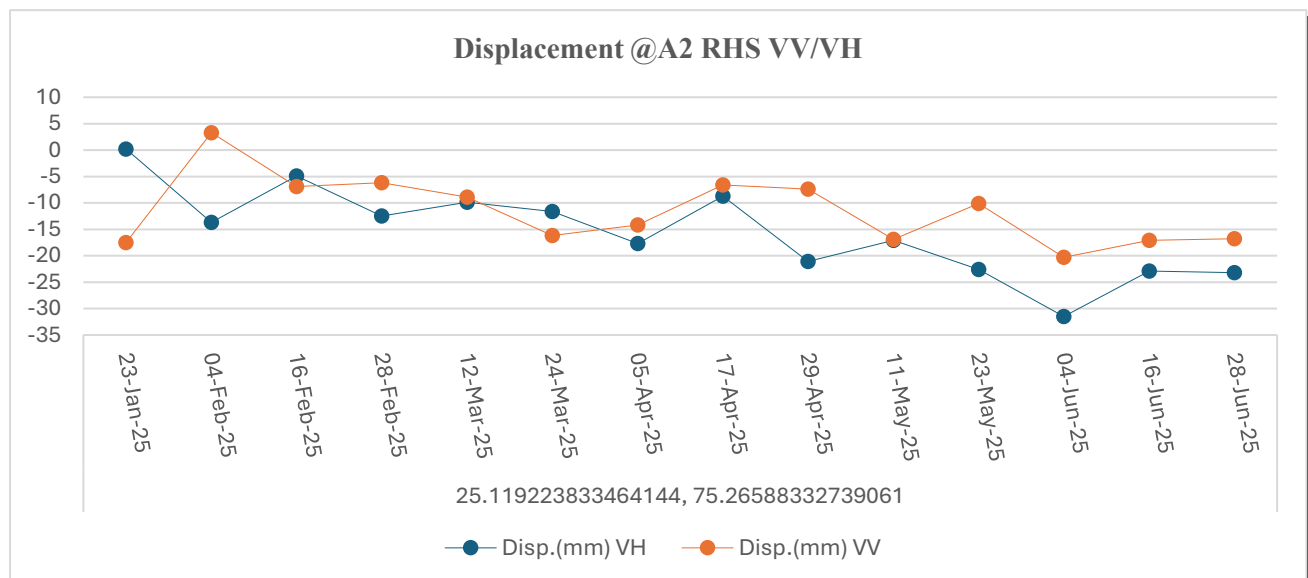
Critical Events & Anomalies:

- **January 2025:** VH begins at around -15 mm, while VV is near 0 mm, indicating initial lateral displacement with negligible vertical response.
- **June 2025:** VV peaks at approximately +22 mm, while VH stays negative, confirming persistent lateral displacement combined with increasing vertical uplift toward the end of monitoring.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.119223833464144, 75.26588332739061)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Horizontal displacement at A2 RHS shows strong negative dominance with high variability, ranging approximately from 0 mm to –32 mm.
- **Horizontal (VH) displacement** Horizontal displacement at A2 RHS shows strong negative dominance with high variability, ranging approximately from 0 mm to –32 mm.

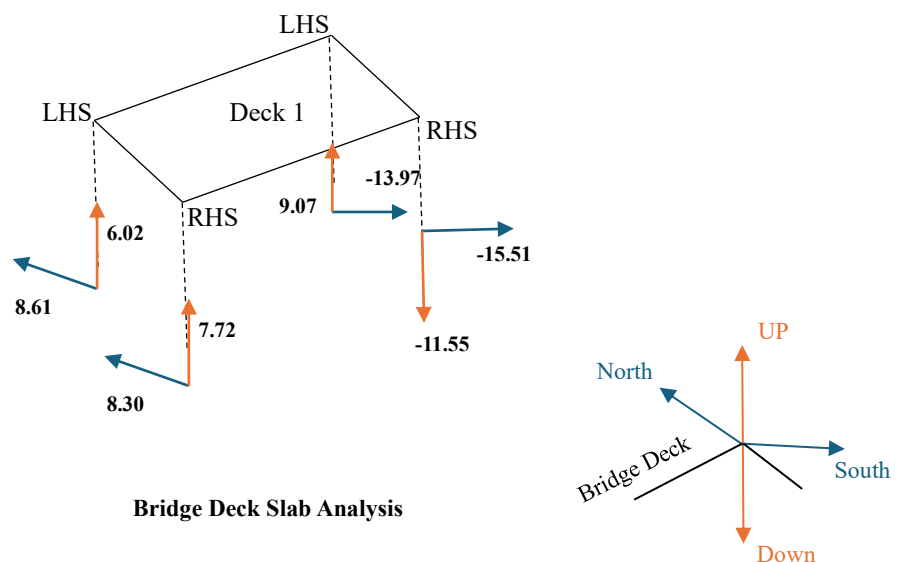
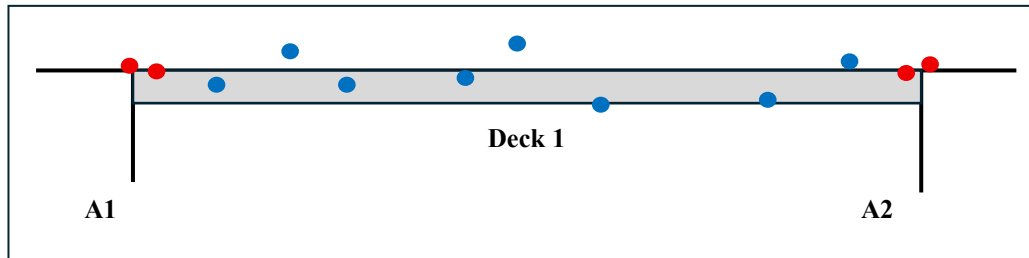
Critical Events & Anomalies:

- **January 2025:** VH starts near 0 mm, while VV is already at –18 mm, indicating early vertical settlement with minimal horizontal movement.
- **May 2025:** VH deteriorates sharply, reaching –22 mm to –25 mm, while VV fluctuates between –10 mm and –15 mm, indicating renewed and intensified horizontal instability.
- **June 2025:** VH partially recovers to –22 mm, while VV stabilises near –18 mm, indicating persistent residual deformation with no full recovery.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits strongly non-uniform vertical displacement, indicating significant differential vertical response between the LHS and RHS regions. The coexistence of large uplift on both supports and significant settlement on the RHS confirms vertical rotation combined with localized bending, rather than uniform vertical deflection.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates strong directional variability across the deck, reflecting a torsion-dominated deformation pattern.

CONCLUSION:

The bridge deck slab demonstrates a torsion-controlled deformation response, characterized by large asymmetric horizontal displacement combined with significant vertical uplift and settlement. Overall, the magnitude and asymmetry of the observed displacements indicate structurally significant behaviour, warranting continued monitoring and detailed assessment, particularly with respect to torsional effects and support performance